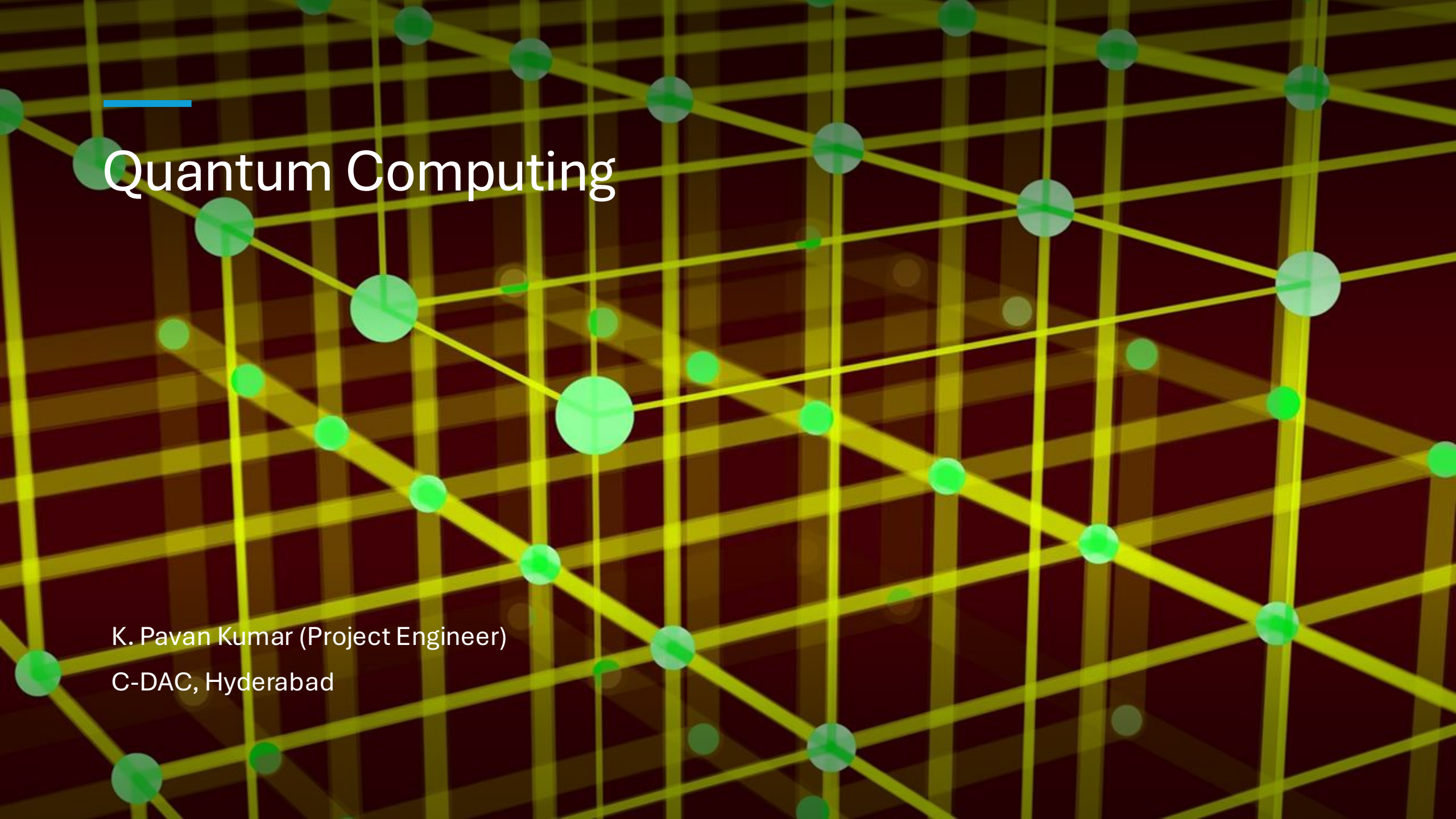
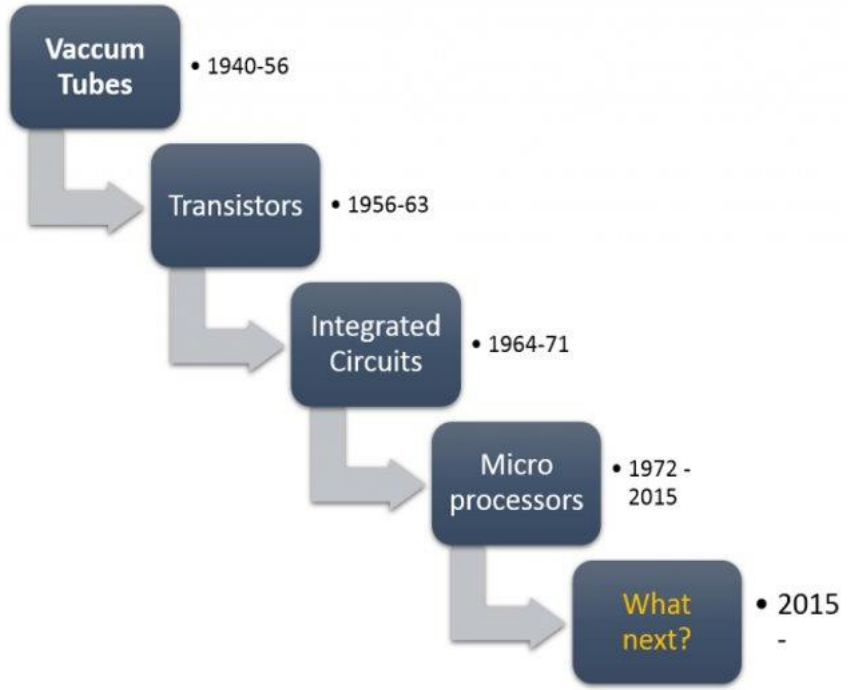


Quantum Computing

K. Pavan Kumar (Project Engineer)

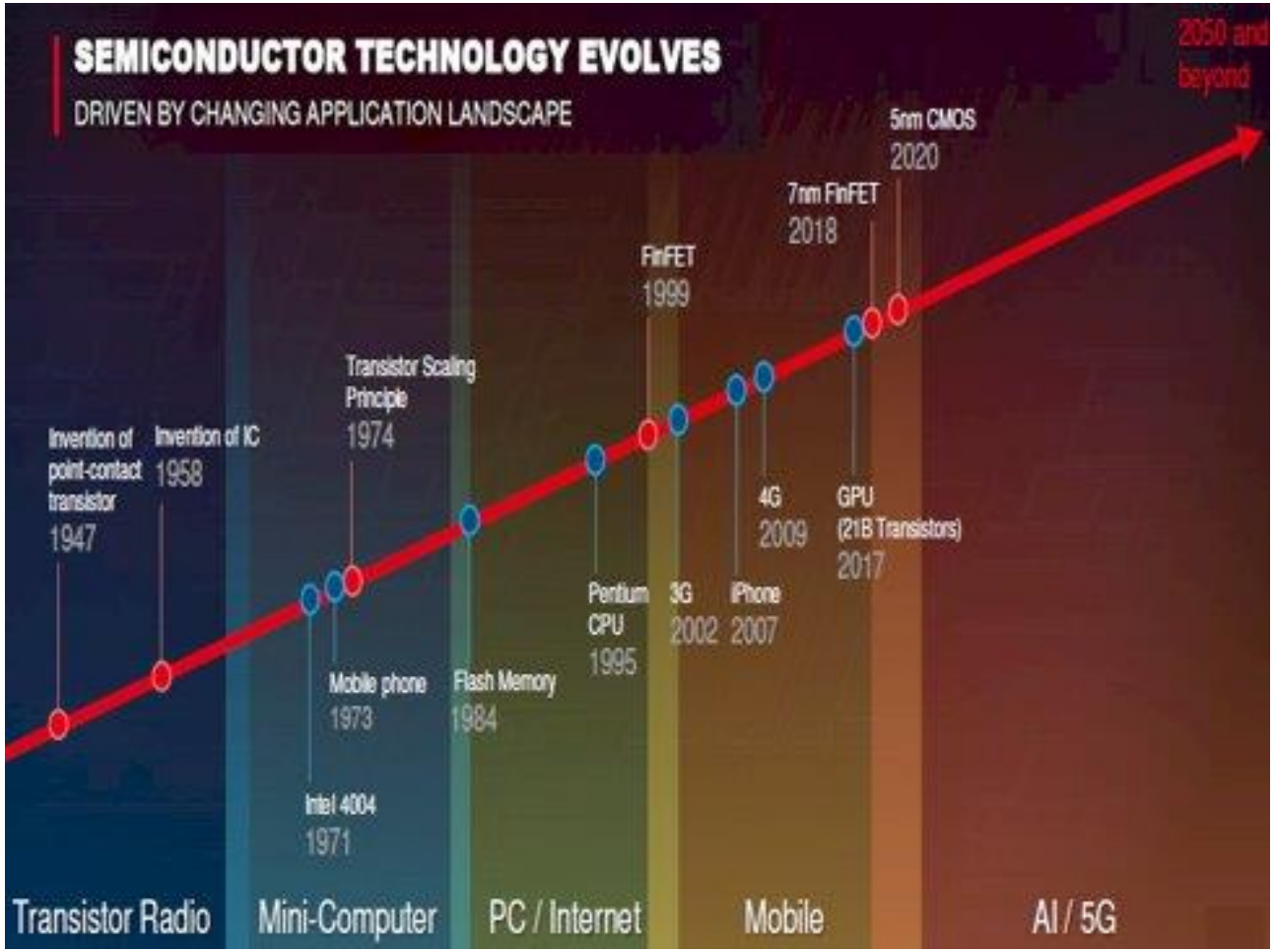
C-DAC, Hyderabad

Evolution of Computers:



Transistor based Computing:

- **Moore's Law** states that the number of transistors on a chip roughly doubles every 18–24 months, leading to continuous performance improvement.
- Today, Moore's Law is slowing down due to physical limits like **heat dissipation**, **leakage current**, and **atomic-scale transistor sizes**.
- Many modern problems (optimization, molecular simulation, cryptography) grow exponentially and are **inefficient** for classical computers.
- ***Quantum computing*** offers a new paradigm, using qubits, superposition, and entanglement to solve specific problems more efficiently than classical systems.



Introduction to Quantum Technology:

- Quantum technology is a multi-disciplinary field encompassing quantum physics, computer science and engineering
- All electronic and mechanical information processing devices that pervade our modern society, from the fastest supercomputer available to the humble piece of chalk in a teacher's hand, are governed by the laws of classical physics
- Quantum technologies are set to alter this scenario
- As research and development in this field continues, the resulting technology could directly impact how we analyze data, design novel instruments, communicate, travel, conduct financial transactions, and even diagnose and treat illnesses.

Quantum Technologies and Applications Areas

- Quantum computing and quantum simulation
- Quantum communication
- Quantum sensing and metrology
- Quantum materials and device



The beginnings

- In 19th century, people thought classical physics and electromagnetic theory explained everything. Scientists thought there is only precise measurements are left.
- But the nature said no.
- Scientists observed abnormalities in Black body radiation, Rutherford's atomic theory etc.. Where classical calculations failed.
- Scientists observed the hints that there are the behaviours that classical physics could not explain.
- These experimental observations such as photo electric effect, Bohr's atomic model lead to quantum theory.

The beginnings

Then finally we came to the conclusion that sub-atomic particles behave differently when compared to the objects with which they are made of.

Dual nature of matter was observed. A light is a wave (explained through phenomenon like superposition, interference and diffraction). A light is a particle (explained through phenomenon like photo electric effect)

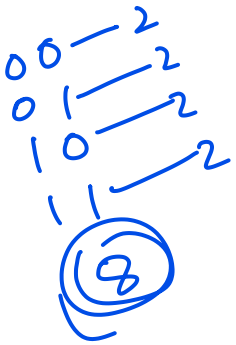
Also scientist observed peculiar behaviour, not only wave-partice duality but also quantum tunnelling and quantum entanglement.

All these principles lead to the idea, famously quoted by Richard Feynman, "Nature is quantum , If we need our computers to simulate nature, then they should be quantum".....

What is quantum computing ?

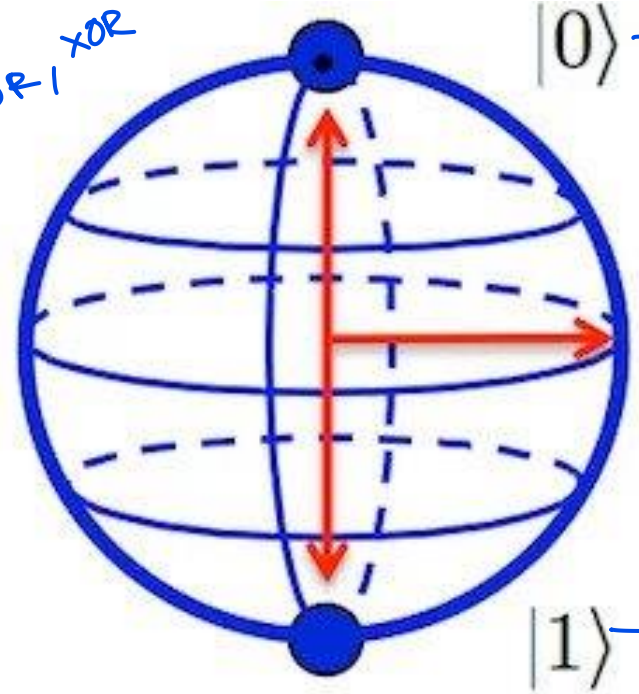
- Quantum computing is a type of nonclassical computing based on the quantum state of subatomic particles.
- The behaviour is governed by the laws of quantum mechanics
- Quantum computing is fundamentally different from classic computers, which operate using binary bits. This means the bits are either 0 or 1, true or false, positive or negative
- However, in quantum computing, the bit is referred to as a quantum bit, or qubit.
- Unlike the strictly binary bits of classic computing, qubits can, strangely, represent a range of values in one qubit. This representation is called “superpositioning.”

Comparison of Bit and Qubit:





AND/OR/XOR



$$|0\rangle \rightarrow \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$|1\rangle \rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|00\rangle = \begin{bmatrix} \frac{|0\rangle+|1\rangle}{\sqrt{2}} \\ \frac{|0\rangle+|1\rangle}{\sqrt{2}} \end{bmatrix}$$

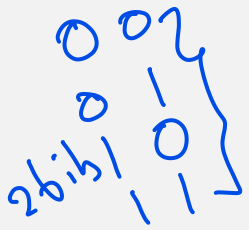
$$= |00\rangle + |01\rangle + |10\rangle + |11\rangle$$

2

Superposition

Classical Bit **Qubit**

Contd..



- **Superposition** is what gives quantum computers speed and parallelism, as each qubit can represent a quantitative solution to a problem.
- Further, qubits can be linked with other qubits in a process called entanglement. When combined with superposition, quantum computers can process a massive number of possible outcomes at the same time.
- The number of high-quality qubits necessary to make a viable quantum computer depends on the problem.
- The ability for a quantum computer to outperform a classical computer is called “quantum supremacy.” While it may sound like a sci-fi dream, experts believe that for a limited number of computing problems, quantum supremacy will be a reality in a matter of years.

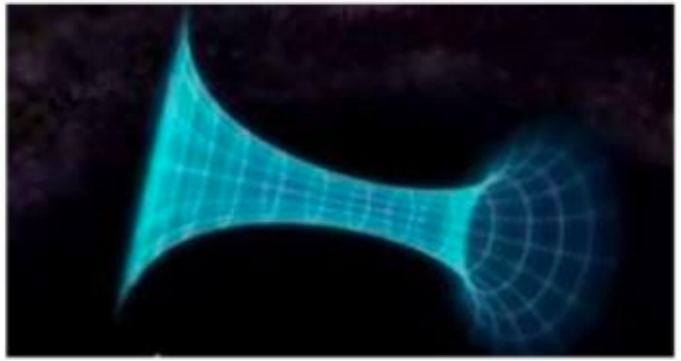


Quantum Mechanics Principles

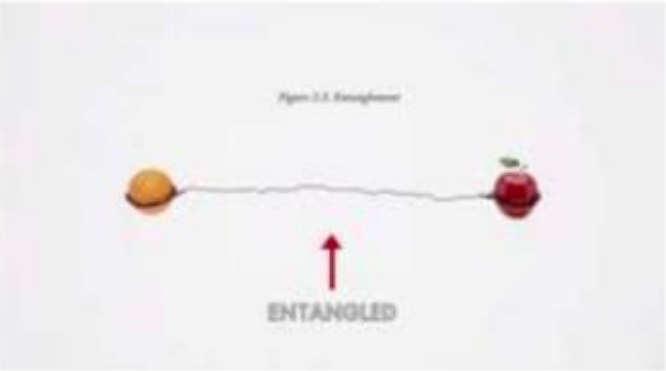
Superposition



Quantum Tunneling



Entanglement



The principle of quantum superposition states that if a physical system may be in one of many configurations then the most general state is a combination of all of these possibilities.

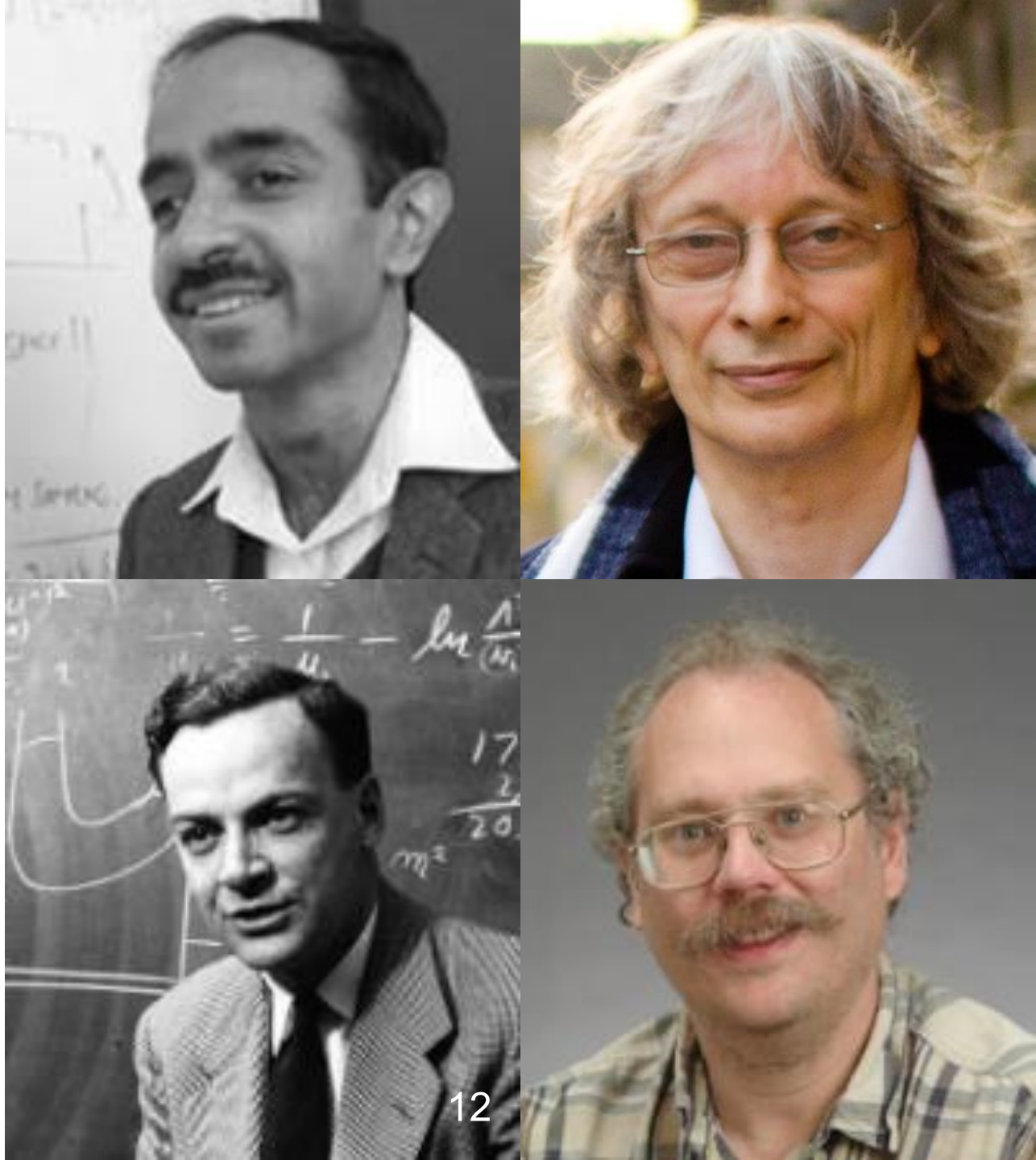
Quantum tunnelling is the quantum mechanical phenomenon where a particle passes through a potential barrier that it classically cannot surmount.

Quantum entanglement is a phenomenon in which the quantum states of two or more objects have to be described with reference to each other, even though the individual objects may be spatially separated.

Courtesy: Wikipedia

Brief History of Quantum Computing

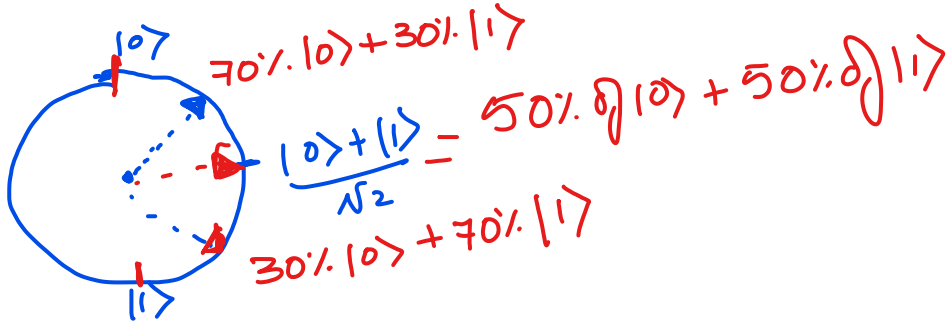
- 1982 - Feynman proposed the idea of creating machines based on the laws of quantum mechanics instead of the laws of classical physics.
- 1985 - David Deutsch developed the quantum Turing machine, showing that quantum circuits are universal.
- 1994 - Peter Shor came up with a quantum algorithm to factor very large numbers in polynomial time.
- 1997 - Lov Grover develops a quantum search algorithm with $O(\sqrt{N})$ complexity



Introduction to Qubits

- A qubit, similar to classical bit can take two values 0 and 1, represend in Bra-Ket notation as $|0\rangle$ and $|1\rangle$
- It is different from the classical bit in its property called superposition, where the laws of quantum mechanics allows the qubit to be in superposition of the states $|0\rangle$ and $|1\rangle$
- All the distinct states of $|0\rangle$ and $|1\rangle$ can be visualized using Bloch sphere of unit radius
- $|0\rangle$ and $|1\rangle$ are called computational basis states, and are represented on the z-axis on the Bloch sphere.
- Similarly there are other basis states such as $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$, $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$ represented on the X-axis of the Bloch sphere
- $|+i\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, |-i\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$ represented on the y-axis of the Bloch sphere

Qubit representation



Qubits are typically represented by “kets”. The vector representation of a single qubit is:

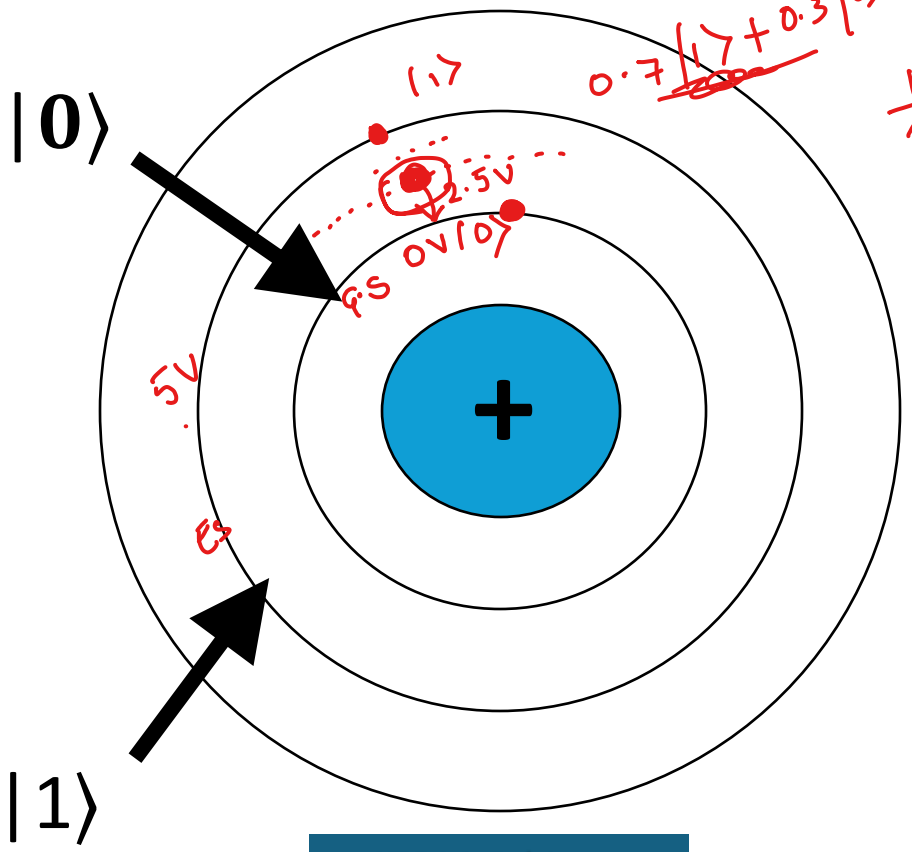
$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

where α and β are the complex probability amplitudes of the qubit which determine the probability of measuring a |0⟩ or a |1⟩, while measuring the state of the qubit. st. α and β are complex numbers satisfying

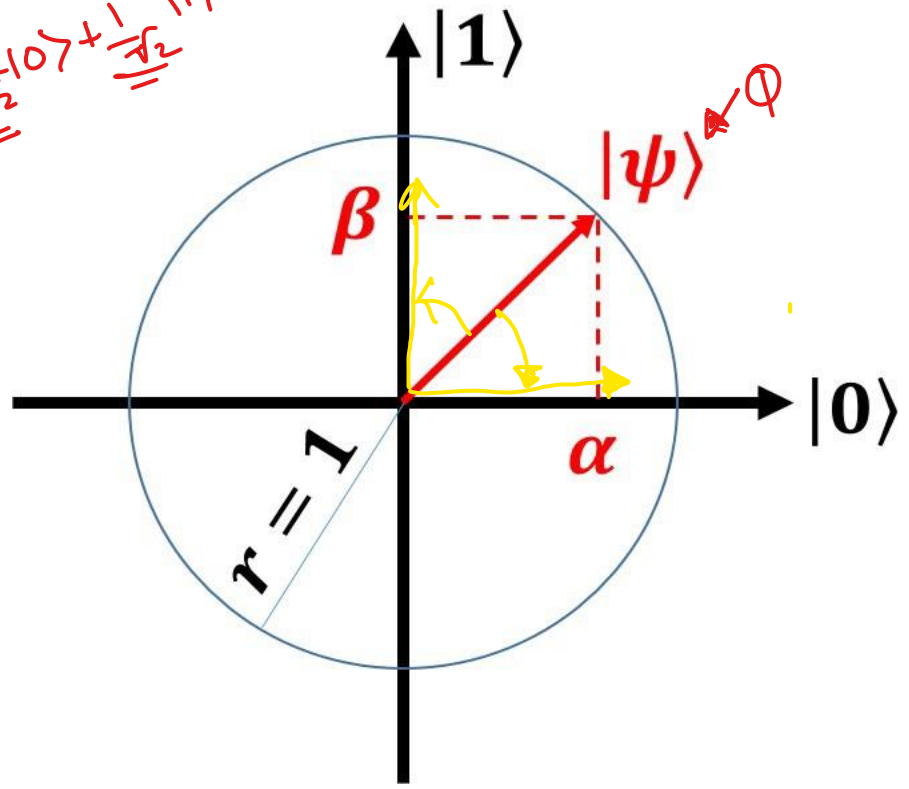
$$|\alpha|^2 + |\beta|^2 = 1$$

Measuring a qubit, whose state given by will yield the classical value of either zero (|0⟩) with probability $|\alpha|^2$ or one (|1⟩) with probability $|\beta|^2$.

Quantum Systems



ATOM Orbits as quantum states



Superposition

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$|\psi\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

How Quantum computing addresses limitations

- **Superposition** Quantum bits (qubits) can exist in multiple states simultaneously, allowing quantum computers to process a vast number of possibilities at once.
- **Entanglement** Qubits can be entangled, meaning the state of one affects the other instantly—enabling faster and more complex computations.
- **Quantum Parallelism** Quantum computers can explore many solutions simultaneously, making them exponentially faster for certain tasks like optimization and cryptography.
- **Efficient Simulation of Quantum Systems** Since quantum computers operate on quantum principles, they're naturally suited to simulate molecules, atoms, and other quantum systems.
- Quantum computing isn't here to replace classical computing entirely—it's more like a powerful new tool for problems that classical systems just can't crack efficiently.

Bloch Sphere

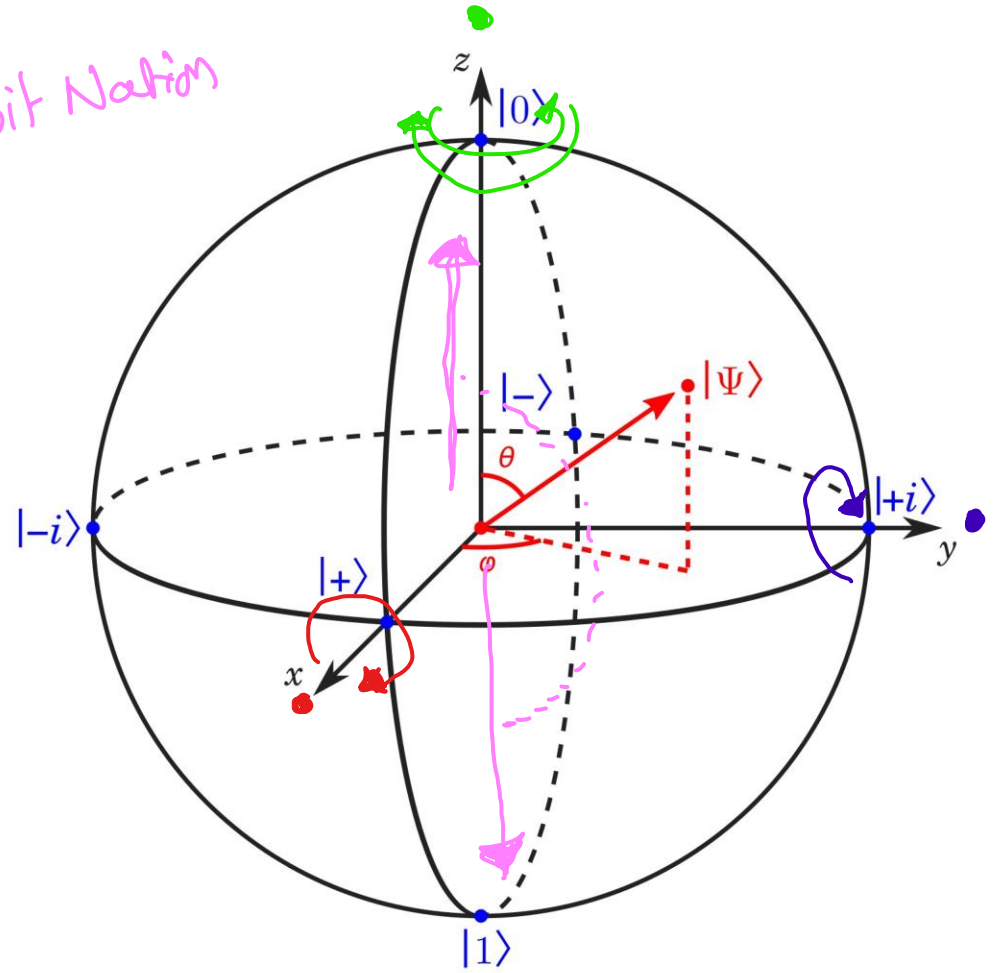
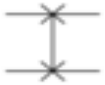
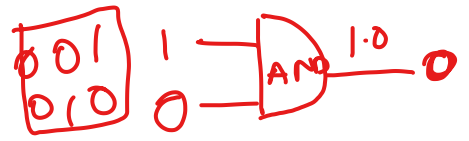
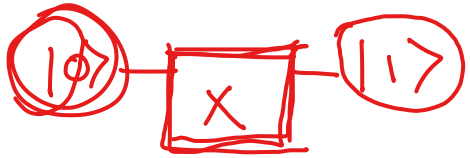


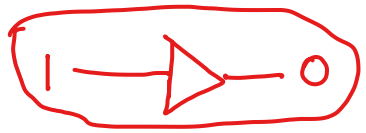
Figure: Bloch Sphere

Image credits: https://prefetch.eu/know/concept/bloch-sphere/

Operator	Gate(s)		Matrix
Pauli-X (X)			$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Pauli-Y (Y)			$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Pauli-Z (Z)			$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
Hadamard (H)			$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
Phase (S, P)			$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
$\pi/8$ (T)			$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
Controlled Not (CNOT, CX)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
Controlled Z (CZ)			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$
SWAP			$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Toffoli (CCNOT, CCX, TOFF)			$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$

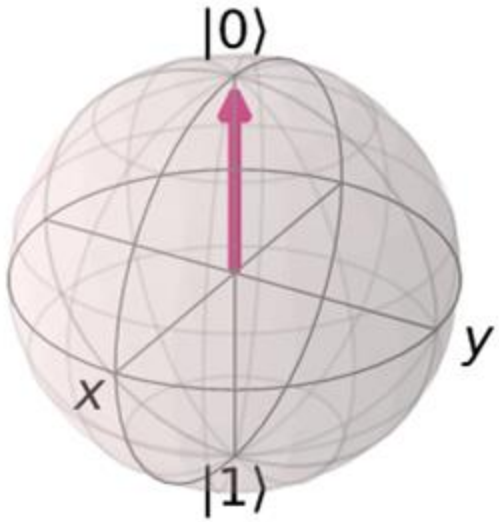




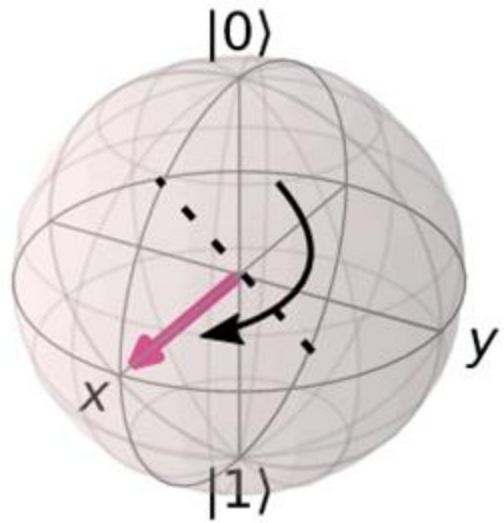


Reversibility Nature.

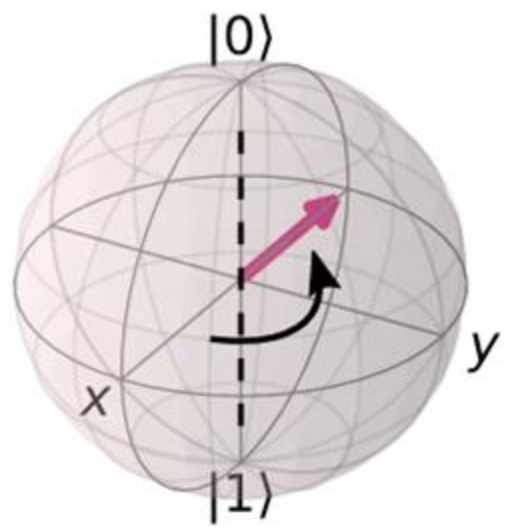
Start



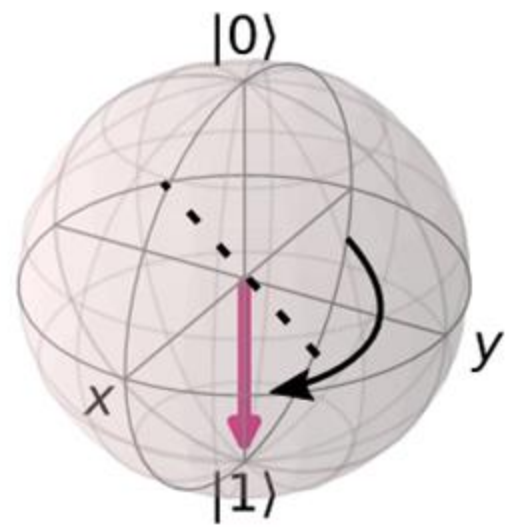
Apply H-gate

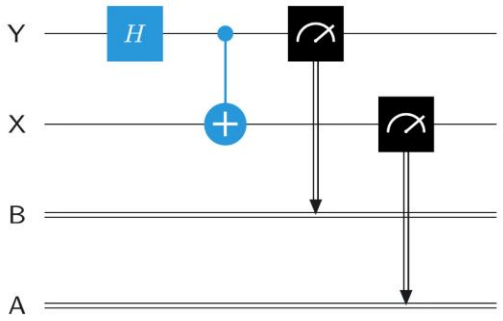


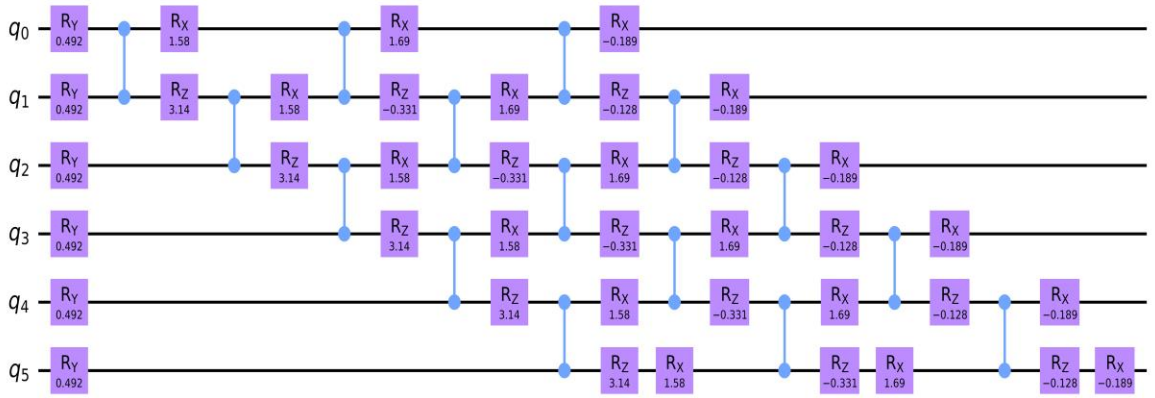
Apply Z-gate



Apply H-gate







Quantum Circuits

Various methods to physically implement Qubits

1. Superconducting qubits	**Superconducting materials, such as niobium, which exhibit zero electrical resistance at low temperatures**
2. Semiconductor Quantum Dots	Semiconductor materials, such as silicon or gallium arsenide
3. Trapped Ion	ions that are confined and manipulated using electromagnetic fields in a vacuum chamber
4. Photonic qubits	Encoding quantum information in the properties of photons
5. Defect-based qubits – NP in Diamond	specific defects within a material, such as the nitrogen-vacancy (NV) centers in diamond
6. Topological nanowire qubits – Majorana Qubits	one-dimensional nanowires that host exotic quasiparticles called Majorana fermions
7. Nuclear Magnetic Resonance	interaction between the magnetic properties of atomic nuclei and an applied magnetic field

Obstacles

There are a number of technical challenges in building a large-scale quantum computer

- Scalable physically to increase the number of qubits
- Qubits that can be initialized to arbitrary values
- Quantum gates that are faster than decoherence time
- Universal gate set
- Qubits that can be read easily
- Sourcing parts for quantum computing

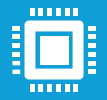
Where do we stand today....



We are in the NISQ era (Noisy Intermediate-Scale Quantum): quantum computers exist but are limited by noise and decoherence.



Quantum advantage is task-specific, demonstrated only for selected problems—not general-purpose computing yet.



Most real-world work today uses hybrid quantum–classical approaches, running quantum algorithms on simulators or cloud-based QPUs.



Research and industry focus on experimentation, error reduction, better algorithms, and preparing for fault-tolerant quantum systems.

Quantum frameworks

Qiskit : IBM's quantum information software kit for simulating, compiling and executing quantum programs.

Cirq : Framework for creating, editing, and invoking Noisy Intermediate Scale Quantum (NISQ) circuits.

Forest Rigetti's software library for writing, simulating, compiling and executing quantum programs.

Q# : Microsoft's quantum programming language with Visual Studio integration

ProjectQ: Hardware-agnostic framework with compiler and simulator with emulation capabilities.

Ocean :D-Wave System's suite of tools for solving hard problems with quantum computers.

XACC: Extreme-scale programming model for quantum acceleration within high-performance computing.

Current Research Trends:



Defense Applications: Research focuses on **quantum-enhanced sensing and communication**, including *quantum radar* for low-observable target detection, **quantum navigation without GPS**, ultra-precise timing, and **secure quantum communication** for resilient defense networks.



Medical & Life Sciences – Quantum-Enhanced Molecular & Biological Modeling: Research explores **quantum simulation of complex biological systems**, including molecular interactions, protein folding, genetic analysis, and **drug discovery**, aiming for faster and more accurate understanding at the atomic scale.



Logistics & Industrial Systems – Quantum Optimization & Decision Making: Focus is on **quantum and hybrid optimization techniques** to solve large-scale scheduling, routing, supply-chain management, and resource allocation problems that are computationally expensive for classical systems.



Cryptography & Cybersecurity – Quantum-Safe Security & Communication: Research targets **quantum-resistant encryption algorithms**, secure key exchange, and **quantum key distribution (QKD)** to protect data against both classical and future quantum attacks.

Research works Carried out in C-DAC

QSim: Quantum Computer Simulator Toolkit

QSim
QC Workbench

- QSim is like a playground for anyone passionate to learn or experiment in quantum computing, be it students, faculty or researchers
- QSim offers a robust QC Simulator integrated with a GUI based Workbench in which you can create quantum circuits and quantum programs, view the outputs, online help, solved examples and related literature/material
- QSim is realistic simulator considering effects of noise. It allows researchers to explore Quantum Algorithms under idealized conditions and help prepare experiments to run on actual Quantum Hardware.
- Provides access to multiple users to simulate their Quantum Circuits



Simulate Quantum Circuits

Quantum Noise

Intuitive UI

User management

Example & Help

Code Editor

PARAM QSim Cloud : Available on cloud using HPC Infrastructure PARAM SUPERCOMPUTE
(developed and deployed under NSM program)

PARAM SHAVAK QSim : Standalone system with Quantum Simulator in a box

QSim: Quantum Computer Simulator Toolkit (qctoolkit.in)

QSim
QC Workbench



Simulate quantum circuits: Simulation of Quantum circuits with custom parameters.



Quantum Noise: Realistic simulator considering effects of noise



Intuitive UI: Intuitive UI/UX helps users to conceptualize and create quantum programs



Examples & Help: Online help, solved examples and learning material.



Secured user management: Secure user management with options to save quantum programs/circuits



Code editor: Advanced Python code editor for Quantum Circuits.

भारतीय विज्ञान संस्थान

HPC based Quantum Accelerators for enabling Quantum Computing on Supercomputers



Overall Objective: The objective is to build a Quantum Accelerators wherein an Accelerator card is integrated with HPC nodes to achieve computational tasks that are otherwise beyond the reach of the available Quantum Simulators. It is proposed to develop three testbed GPU/Vector/FPGA based and enable QSim/other open source simulator environment. Two deployment on Param Siddhi and Param Utkarsh and Cloud based Quantum Experience Centre.

Collaborating Centres: C-DAC Pune, Bengaluru, Hyderabad, Noida, Patna

CDAC Hyderabad

- Porting of QSim and other Open Source Quantum Simulators on different Accelerator Platforms (GPU/VECTOR)
- Carry out Benchmarking on these platforms
- Dissemination and Knowledge Sharing



GPU CARD



VECTOR ENGINE CARD

Development of Quantum Machine Learning Use Cases and Applications



- **Objectives:**
  - ✓ Development of Quantum Machine Learning based applications in the areas of Healthcare, surveillance, geology, cyber security, emergency management, remote sensing and chemistry.
  - ✓ Capacity building in the areas of Quantum Computing and Quantum Machine Learning.

- **Implementing agencies :** CDAC (Hyderabad, Mohali, Noida, CINE, Delhi), IIT (Roorkee, Ropar, Gandhinagar), IIIT Hyderabad

Target Deliverables: Indigenous QML Solutions & Frameworks for

- ✓ Early Disease Detection, Mineral Prospectivity Analysis, Image and Video Compression, Weather Forecast and Climate Modelling, Entity and Anomaly Detection, Escape/Evacuation Routing, Brain Activities Analysis, Malware/Ransomware Analysis, Analysis of Satellite Images, Algorithms for Behavioral Pattern Analysis, Simulation of Molecular Dynamic, Learning Physical Systems and Quantum States
- ✓ Capacity Building: Training of 600 Manpower in the areas of Quantum Machine learning through online and offline programmes

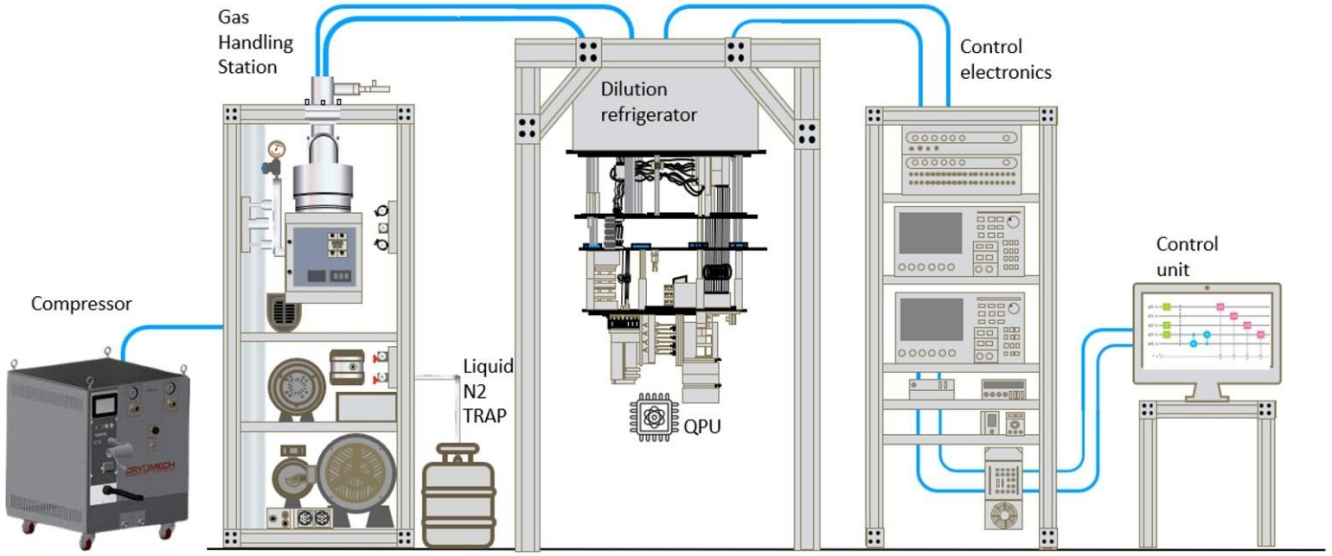
Impact: To Leverage Quantum Advantage in Solving Complex Real-World Problems Using QML and getting ready for Quantum Age challenges along with Capacity Building in this Critical Domain.

Establishment of Integrated Quantum Computing Reference Facility

Objectives:

The main aim of the project is establishment of Integrated Quantum Computer Reference Facility which will serve as a cutting-edge research and development hub, fostering collaborations, advancing quantum computing technology.

The reference facility is a superconducting based Quantum Computer integrated by global partners.



Collaborating Centres: C-DAC Pune and Bengaluru

Benchmarking of Quantum Simulators:

- Benchmarked performance metrics of various popular Quantum Algorithms like, Deutsch-Jozsa, Grover's Search, QFT on various frameworks like Qiskit, Cirq, Bra-Ket, QSim.
- Benchmarks were performed on various Hardware configurations like CPU, GPU, Vector Accelerators, FPGA.
- Testing was performed for both Ideal and Noisy conditions, to check, how well the simulators are performing w.r.t Hardware limitations.

Configuration	CPU	GPU	Vector Card
Device	12th Gen Intel® Core™ i9-12900	NVIDIA A100-SXM4 40GB	NEC SX-Aurora Tsubasa VE20B
Device Cores	16 Cores (24 Threads)	6912 CUDA Cores	8 Vector Cores
Memory	64 GB	2 × 40 GB	48 GB
Host CPU	12th Gen Intel® Core™ i9-12900	AMD EPYC 7742, 64-Core	Intel Xeon Gold 6326 @ 2.90 GHz
Host CPU Cores	16 Cores	128 Cores	64 Cores

Metrics Used for Quantum Benchmarking:

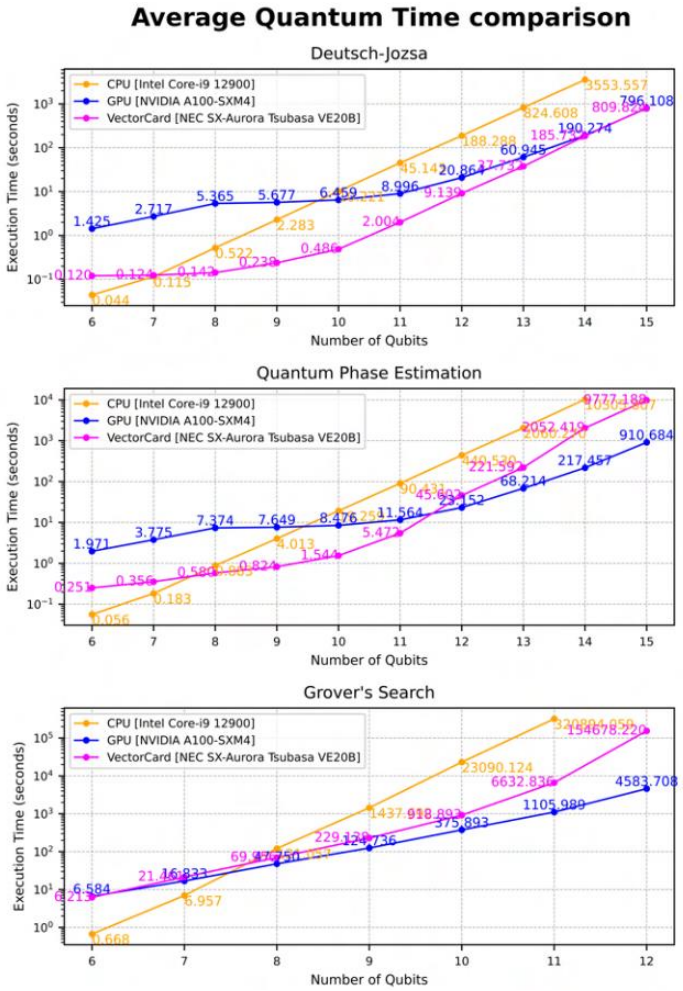
1. **Circuit Metrics:** Gate count and circuit depth → indicate **hardware complexity and scalability**.
2. **Execution Metrics:** Execution time and backend constraints → reflect **practical feasibility**.
3. **Fidelity Metrics:** Polarization fidelity and Hellinger fidelity → measure **output correctness under noise**.
4. **Comparative Analysis:** Results are stored in structured datasets (CSV) to enable **cross-backend and cross-benchmark comparison**.

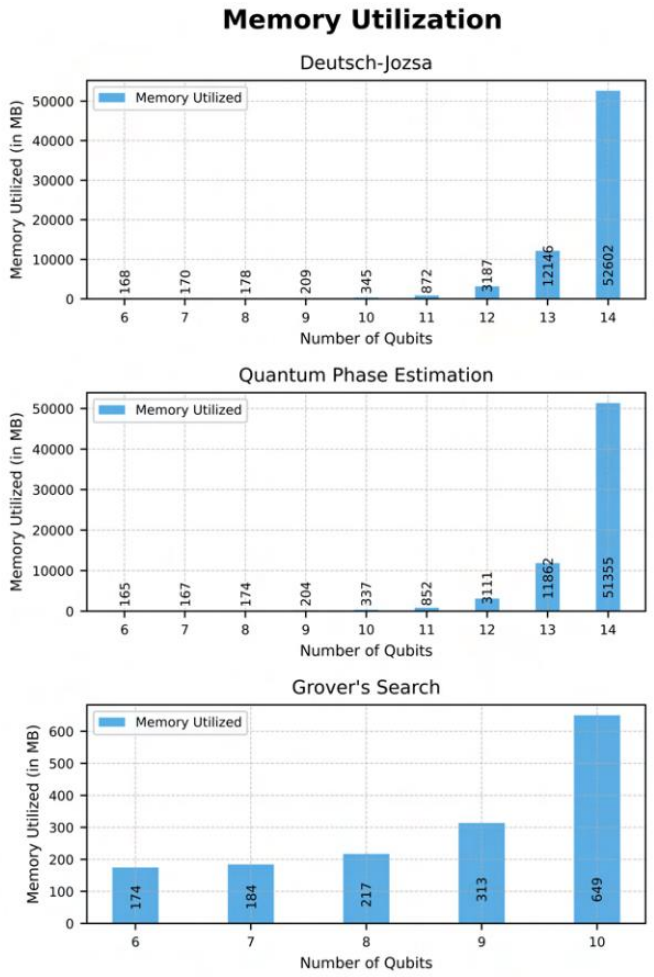
Key Insight: Benchmarking helps identify **which quantum backends are suitable for which types of algorithms**.

Benchmarking Results

Reference:

[Performance Evaluation of an Indigenous Density Matrix Quantum Computer Simulator for Scalable Workloads]





QAOA Enhancement for Classical ML and DL Pipelines:

Hybrid quantum–classical approach where QAOA is used as a subroutine within classical ML/DL models.

QAOA helps solve **combinatorial optimization sub-problems** such as feature selection, hyperparameter tuning, and clustering.

Classical neural networks handle **data representation and learning**, while QAOA provides **quantum-assisted optimization**.

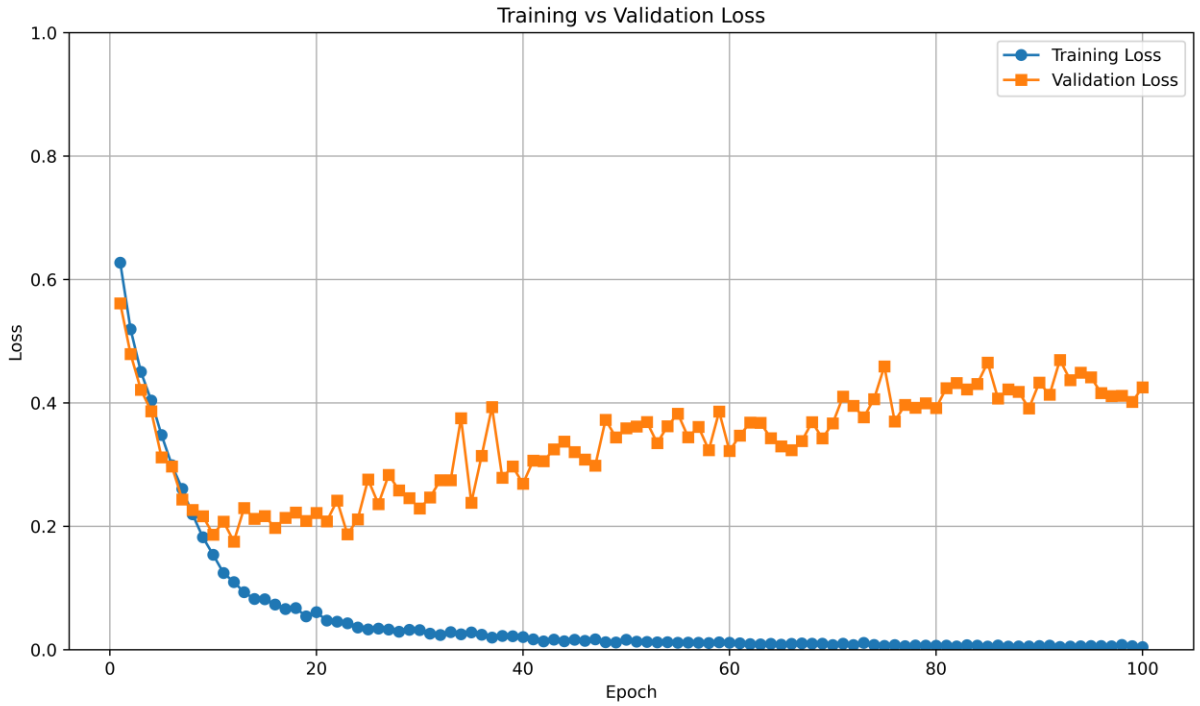
This integration aims to improve **solution quality and convergence**, especially for hard optimization tasks.

Improvements in CNN Loss Minimization:

With QAOA- Enhancement



Without QAOA-Enhancement




Thank you